
🚀 Increment Expression

> The **increment expression** increases a variable's value by 1.

- Syntax: $x++$
- Equivalent to: $x = x + 1$
- Example flow:
  1. Set x = 5
  2. Apply $x++$ → x becomes **6**

🔽 Decrement Expression

> The **decrement expression** decreases a variable's value by 1.

- Syntax: $x--$
- Equivalent to: $x = x - 1$
- Example flow:
  1. Starting value x = 5
  2. Apply $x--$ → x becomes **4**

😄 C++ Joke

> The language name **C++** is a playful reference to an "increment" of the C language, i.e., **C + 1**.

- Highlights that ++ and -- always change a value by **exactly one**.

📚 Compound Assignment Operators

> **Compound assignment operators** combine an arithmetic operation with assignment, providing a concise way to update a variable.

Operator	Equivalent Expression	Example Result

$+=$	$x = x + 2$ (or $x = x + y$)	If x = 5, $x+=2$ → **7**; if y = 1, $x+=y$ → **8**
$-=$	$x = x - 2$	–
$*=$	$x = x * 2$	–
$/=$	$x = x/2$	–

- These operators work with any numeric type and follow the same precedence rules as their non-compound counterparts.

❓ Quiz Prompt

> A brief multiple-choice quiz follows to test understanding of increment, decrement, and compound assignment concepts.